

Math Thermo
class # 07
02/10/2026

Legendre Transform

If f is convex, for all $p \in \text{range}(f')$,

$$f^*(p) = x_p \cdot p - f(x_p),$$

where

$$f'(x_p) = p.$$

This transforms the function $f(x)$, a function of x , to $f^*(p)$, a function of p , the derivative variable. Recall f^* has all the information of f !

In Thermodynamics we use the negative Legendre Transform, but the idea is the same. We want to transform to an equivalent function where a derivative replaces one of the original variables.

Recall $\tilde{U} = \tilde{U}(S, V, N)$ is convex. We want to transform to an equivalent function $\tilde{F} = \tilde{F}(T, V, N)$, where the derivative $T = \frac{\partial \tilde{U}}{\partial S}$ replaces S .

Recall,
$$\tilde{F}(T, V, N) = \tilde{U}(S_T, V, N) - TS_T,$$

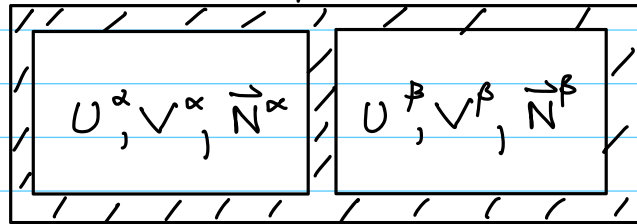
where $S_T = S_T(V, N)$ is the unique solution of the equation

$$T_U(S_T(V, N), V, N) = T.$$

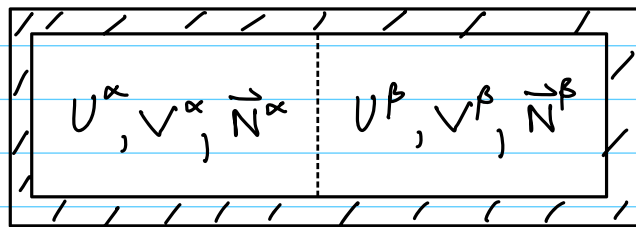
Note that $\tilde{U}(S, V, N)$ and $\tilde{F}(T, V, N)$ are not the same, but they have the same information content.

Equilibrium with a Heat Bath

Consider two isolated systems that are initially isolated from each other.



Now, let us remove the isolating wall so that energy, volume, and mass may be exchanged



The entropy will attain its maximum at equilibrium and

$$\begin{aligned} T^\alpha &= T^\beta \\ P^\alpha &= P^\beta \\ \mu_i^\alpha &= \mu_i^\beta, \quad i = 1, \dots, r. \end{aligned}$$

Now, suppose that instead of isolating from the universe, our composite system is put into contact with a heat bath.

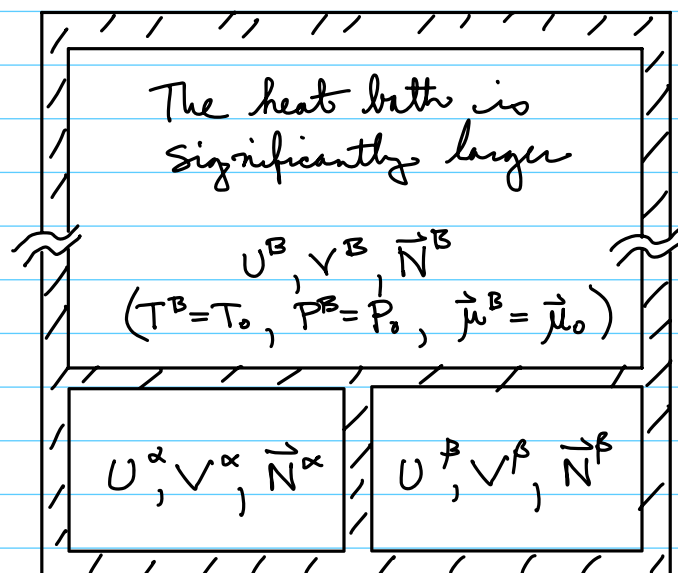
Definition (7.1): A heat bath is an otherwise-isolated thermodynamic system that is so large that, when it exchanges a finite amount of energy with an ordinary, otherwise-isolated, compound system its temperature, pressure, and chemical potential changes are so small as to be negligible.

Similarly, we have the following

heat-pressure bath: finite exchanges of energy and volume lead to negligible changes in T , P , and μ in bath.

heat-pressure-chemical bath: finite exchanges of energy, volume, and matter lead to negligible changes in T , P , and μ in the bath.

let us consider a picture for the case of the heat bath.



let us recall the "integrated form" of the entropy functions.

$$\begin{aligned}\tilde{S}^\square &= \tilde{S}^\square(U^\square, V^\square, \vec{N}^\square) \\ &= \frac{1}{T^\square} U^\square + \frac{P^\square}{T^\square} V^\square - \frac{\vec{\mu}^\square}{T^\square} \cdot \vec{N}^\square\end{aligned}$$

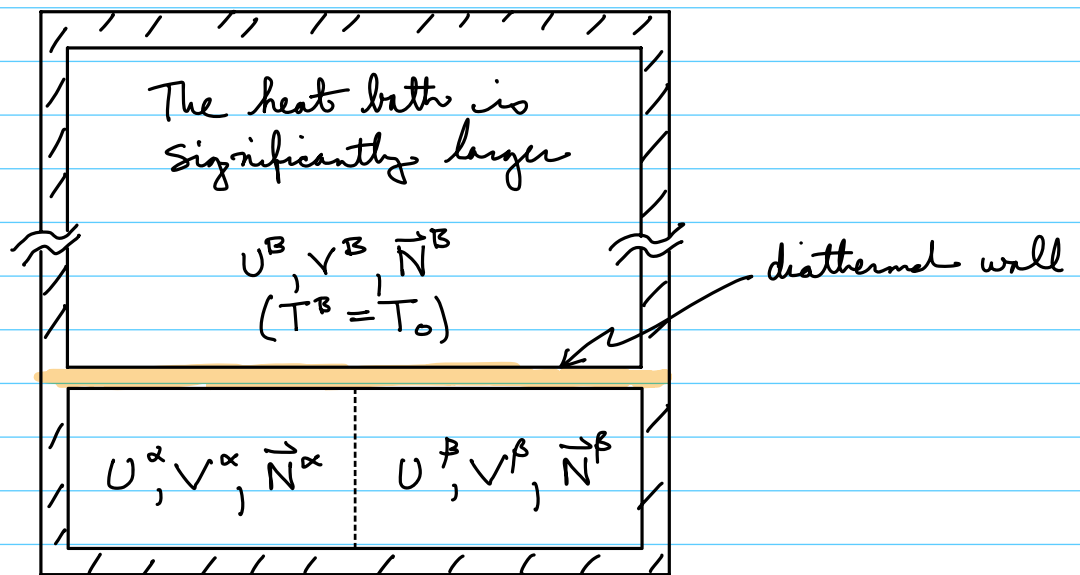
where

$$T^\square = T^\square(U^\square, V^\square, \vec{N}^\square)$$

et cetera. We will write

$$\begin{aligned}S^{\text{Tot}} &= \tilde{S}^B + \tilde{S}^\alpha + \tilde{S}^\beta \\ &= \tilde{S}^B + \tilde{S}^{\alpha+\beta}\end{aligned}$$

Now, our system will not exchange volume or mass with the bath, only energy, because we will replace the horizontal isolation wall with a diathermal wall.



We know that, at equilibrium

$$T_{eq}^{\alpha} = T_{eq}^{\beta} = T_{eq}^B.$$

Recall the deviation of T_{eq}^B from T_0 at equilibrium is assumed negligible.

Systems α , β , and B can exchange energy. Systems α and β can exchange volume and mass with each other but not with the bath.

Thus,

$$S^{Tot,eq} = \max_C \{ \tilde{S}^B + \tilde{S}^{\alpha+\beta} \}$$

where C is the constraints set,

$$C := \begin{cases} U^{\alpha} + U^{\beta} + U^B = U_0, \\ V^{\alpha} + V^{\beta} = V_0, \\ N_i^{\alpha} + N_i^{\beta} = N_{i,0}. \end{cases}$$

Now

$$\begin{aligned} \tilde{S}^B &= \frac{1}{T_0} U^B + \underbrace{\frac{P_0}{T_0} V^B + \frac{\vec{\mu}_0}{T_0} \cdot \vec{N}^B}_{= \text{const}, C_1} \\ &= \frac{1}{T_0} U^B + C_1. \end{aligned}$$

The change in T^B , P^B , $\vec{\mu}^B$ from T_0 , P_0 , $\vec{\mu}_0$ is negligible because the bath is so large.

Writing

$$U^B = U_0 - U^{\alpha} - U^{\beta}$$

$$\begin{aligned}\tilde{S}^B &= -\frac{U^\alpha}{T_0} - \frac{U^\beta}{T_0} + \frac{U_0}{T_0} + C_1 \\ &= -\frac{U^\alpha}{T_0} - \frac{U^\beta}{T_0} + C_2.\end{aligned}$$

Thus,

$$\begin{aligned}J^{\text{Tot,eq}} &= \max_C \left\{ -\frac{U^\alpha}{T_0} - \frac{U^\beta}{T_0} + C_2 + \tilde{S}^{\alpha+\beta} \right\} \\ &= C_2 + \max_C \left\{ -\frac{(U^\alpha + U^\beta)}{T_0} + \tilde{S}^\alpha + \tilde{S}^\beta \right\} \\ &= C_2 + \max_C \left\{ -\frac{(F^\alpha + F^\beta)}{T_0} \right\} \\ &= C_2 - \frac{1}{T_0} \min_{\substack{V^\alpha + V^\beta = V_0 \\ N_i^\alpha + N_i^\beta = N_{i,0}}} \left\{ F^\alpha(T_0, V^\alpha, \vec{N}^\alpha) + F^\beta(T_0, V^\beta, \vec{N}^\beta) \right\}\end{aligned}$$

Postulate VI: Consider a compound, otherwise-isolated thermodynamic system in contact with an otherwise-isolated heat bath at temperature T_0 . Equilibrium of the compound system is the state satisfying the isothermal condition

$$T^\alpha = T^\beta = T_0$$

which minimizes

$$F^\alpha(T_0, V^\alpha, \vec{N}^\alpha) + F^\beta(T_0, V^\beta, \vec{N}^\beta)$$

Subject to the constraints

$$V^\alpha + V^\beta = V_0, \quad (\text{volume conservation})$$

$$N_i^\alpha + N_i^\beta = N_{i,0}. \quad (\text{mass conservation})$$

Remark: Since the system $\alpha + \beta$ exchanges energy with the heat bath, energy is not conserved in $\alpha + \beta$!

Energy is conserved in the system $B + \alpha + \beta$.

Density Formulations

Suppose that $\tilde{S}$ is the fundamental entropy relation for an isolated system containing a unary ($r=1$) material.

$$\tilde{S} = \tilde{S}(U, V, N).$$

Suppose that $V > 0$ is fixed. Define

$$u := \frac{U}{V}, \quad \text{energy density,}$$

$$\rho := \frac{N}{V}, \quad \text{mole number density,}$$

Then, since, for any $\lambda > 0$,

$$\lambda \tilde{S}(U, V, N) = \tilde{S}(\lambda U, \lambda V, \lambda N),$$

it follows that

$$\begin{aligned}\frac{1}{V} \tilde{S}(U, V, N) &= \tilde{S}\left(\frac{U}{V}, \frac{V}{V}, \frac{N}{V}\right) \\ &= \tilde{S}(u, 1, \rho).\end{aligned}$$

Now, define the entropy density relation

$$\tilde{s} = \tilde{s}(u, \rho) := \tilde{S}(u, 1, \rho).$$

It then follows that

$$\tilde{S}(U, V, N) = V \tilde{s}(u, \rho).$$

likewise, if we define

$$s = \frac{S}{V} \quad \rho = \frac{N}{V}$$

and

$$\tilde{u} = \tilde{u}(s, \rho) := \tilde{U}(s, 1, \rho),$$

then it follows that

$$\tilde{U}(S, V, N) = V \tilde{u}(s, \rho).$$

Example (7.2) Suppose that

$$\tilde{S}(U, V, N) = \left(\frac{NVU R^2}{v_0 \Theta} \right)^{1/3},$$

where $R, v_0, \Theta > 0$ are constants.

Then,

$$\begin{aligned}\tilde{S}(u, \rho) &= \tilde{S}(u, 1, \rho) \\ &= \left(\frac{\rho u R^2}{v_0 \theta} \right)^{1/3}.\end{aligned}$$

$$\begin{aligned}v \tilde{S}(u, \rho) &= \left(\frac{v^3 \rho u R^2}{v_0 \theta} \right)^{1/3} \\ &= \left(\frac{N v U R^2}{v_0 \theta} \right)^{1/3} \\ &= \tilde{S}(U, v, N).\end{aligned}$$

Recall that, for all $\lambda > 0$,

$$T_0(S, v, N) = T_0(\lambda S, \lambda v, \lambda N).$$

Thus,

$$T_0(S, v, N) = T_0(s, 1, \rho).$$

As before, we abuse notation and write

$$T_0(s, \rho) := T_0(s, 1, \rho).$$

Euler and Gibbs - Duhem Equations: Density Forms

For a unary material ($r=1$), recall

$$\tilde{U}(S, V, N) = T_0(S, V, N)S - P_0(S, V, N)V + \mu_0(S, V, N)N$$

Suppose $V > 0$ is fixed. Then,

$$\frac{1}{V} \tilde{U}(S, V, N) = T_0(S, V, N) \frac{S}{V} - P_0(S, V, N) + \mu_0(S, V, N) \frac{N}{V}$$

or, equivalently,

(7.1)

$$\tilde{u}(s, p) = T_0(s, p) s - P_0(s, p) + \mu_0(s, p) p.$$

This is the density version of Euler's equation.

How do we get the Gibbs-Duhem equation?

Let $\vec{\gamma}(r) = (s(r), p(r))$ be a path in density state space. Then

$$\frac{d}{dr} \tilde{u}(\vec{\gamma}(r)) = \frac{\partial \tilde{u}}{\partial s}(\vec{\gamma}(r)) s'(r) + \frac{\partial \tilde{u}}{\partial p}(\vec{\gamma}(r)) p'(r).$$

But

$$\frac{\partial \tilde{u}}{\partial s}(\vec{\gamma}(r)) = T_0(s(r), p(r))$$

and

$$\frac{\partial \tilde{u}}{\partial p}(\vec{\gamma}(r)) = \mu_0(s(r), p(r)).$$

Thus, the combined First and Second laws are

(7.2)

$$\frac{d}{dr} \tilde{u}(\vec{r}(r)) = T_v(\vec{r}(r)) s'(r) + \mu_v(\vec{r}(r)) \rho'(r).$$

On the other hand, differentiating the Euler equation (7.1), we have

(7.3)

$$\begin{aligned} \frac{d}{dr} \tilde{u}(\vec{r}(r)) &= \frac{d}{dr} T_v(\vec{r}(r)) s(r) + T_v(\vec{r}(r)) s'(r) \\ &\quad - \frac{d}{dr} P_v(\vec{r}(r)) + \frac{d}{dr} \mu_v(\vec{r}(r)) \rho(r) \\ &\quad + \mu_v(\vec{r}(r)) \rho'(r). \end{aligned}$$

Therefore, the Gibbs - Duhem equation in density form is

(7.4)

$$0 = \frac{d}{dr} T_v(\vec{r}(r)) s(r) - \frac{d}{dr} P_v(\vec{r}(r)) + \frac{d}{dr} \mu_v(\vec{r}(r)).$$

The entropy density form of the Euler and Gibbs Duhem equations are, as one would expect,

(7.6)

$$\tilde{s}(u, \rho) = \frac{1}{T_s(u, v)} u + \frac{P_s(u, v)}{T_s(u, v)} - \frac{\mu_s(u, v)}{T_s(u, v)} \rho.$$

and

(7.7)

$$\begin{aligned} 0 &= \frac{d}{dr} \left(\frac{1}{T_s(\vec{r}(r))} \right) u(r) + \frac{d}{dr} \left(\frac{P_s(\vec{r}(r))}{T_s(\vec{r}(r))} \right) \\ &\quad - \frac{d}{dr} \left(\frac{\mu_s(\vec{r}(r))}{T_s(\vec{r}(r))} \right) \rho(r). \end{aligned}$$

The density representation of the First and Second laws is

$$(7.8) \quad \frac{d}{dr} (\tilde{S}(\vec{\gamma}(r))) = \frac{1}{T_S(\vec{\gamma}(r))} u'(r) - \frac{\mu_S(\vec{\gamma}(r))}{T_S(\vec{\gamma}(r))} p'(r)$$

where $\vec{\gamma}(r) = (u(r), p(r))$ is a process path in density state space.

let us now define the free energy density.

Recall that

$$F(T, V, \vec{N}) = \tilde{E}(S_T, V, \vec{N}) - T S_T$$

where $S_T = S_T(V, \vec{N})$ is the unique solution of the equation
 $T_U(S_T(V, \vec{N}), V, \vec{N}) = T.$

Theorem (7.3): For any $\lambda > 0$,

$$(7.9) \quad F(T, \lambda V, \lambda \vec{N}) = \lambda F(T, V, \vec{N}).$$

Proof: Since E is homogeneous of degree one and temperature is homogeneous of degree zero, it follows that, for $\lambda > 0$,

$$(7.10) \quad \tilde{E}(\lambda S, \lambda V, \lambda \vec{N}) = \lambda \tilde{E}(S, V, \vec{N})$$

$$(7.11) \quad T_U(\lambda S, \lambda V, \lambda \vec{N}) = T_U(S, V, \vec{N})$$

By definition,

$$(7.12) \quad F(T, \lambda V, \lambda \vec{N}) = \tilde{E}(S_T, \lambda V, \lambda \vec{N}) - TS_T$$

where

$$S_T = S_T(\lambda V, \lambda \vec{N})$$

is the unique solution to

$$(7.13) \quad \begin{aligned} T &= T_U(S_T(\lambda V, \lambda \vec{N}), \lambda V, \lambda \vec{N}) \\ &= T_U\left(\lambda \frac{S_T(\lambda V, \lambda \vec{N})}{\lambda}, \lambda V, \lambda \vec{N}\right) \\ &\stackrel{(7.11)}{=} T_U\left(\frac{S_T(\lambda V, \lambda \vec{N})}{\lambda}, V, \vec{N}\right). \end{aligned}$$

By uniqueness, it must be

$$(7.14) \quad S_T(V, \vec{N}) = \frac{S_T(\lambda V, \lambda \vec{N})}{\lambda},$$

for any $\lambda > 0$. Likewise

$$(7.15) \quad \begin{aligned} \tilde{E}(S_T(\lambda V, \lambda \vec{N}), \lambda V, \lambda \vec{N}) &= \tilde{E}\left(\lambda \frac{S_T(\lambda V, \lambda \vec{N})}{\lambda}, \lambda V, \lambda \vec{N}\right) \\ &\stackrel{(7.10)}{=} \lambda \tilde{E}\left(\frac{S_T(\lambda V, \lambda \vec{N})}{\lambda}, V, \vec{N}\right) \\ &\stackrel{(7.14)}{=} \lambda \tilde{E}(S_T(V, \vec{N}), V, \vec{N}). \end{aligned}$$

Using (7.12),

$$\begin{aligned} F(T, \lambda V, \lambda N) &\stackrel{(7.15)}{=} \lambda \tilde{E}(S_T(V, \vec{N}), V, \vec{N}) \\ &\quad - TS_T(\lambda V, \lambda \vec{N}) \\ &\stackrel{(7.14)}{=} \lambda \tilde{E}(S_T(V, \vec{N}), V, \vec{N}) \\ &\quad - \lambda TS_T(V, \vec{N}) \\ &= \lambda F(T, V, \vec{N}). \quad /// \end{aligned}$$

Free Energy Density

Suppose that $V > 0$ is fixed. Define, as before,

$$\rho := \frac{N}{V}, \quad \text{mole number density}$$

Then, since, for any $\lambda > 0$,

$$\lambda F(T, V, N) = F(T, \lambda V, \lambda N),$$

it follows that

$$\begin{aligned} \frac{1}{V} F(T, V, N) &= F\left(T, \frac{V}{V}, \frac{N}{V}\right) \\ &= F(T, 1, \rho). \end{aligned}$$

Now, define the free energy density

$$f = f(T, p) := F(T, 1, p).$$

It then follows that

$$F(T, V, N) = V f(T, p).$$

More about this next time!